Know: The full Schiffmann algebra $\mathcal{E}$ is constructed as the Drinfeld double of a subalgebra of the Hall algebra of coherent sheaves on an elliptic curve.

This subalgebra corresponds to the subalgebra $\mathcal{E}^{\geq}$ of $\mathcal{E}$ generated by $\mathcal{E}^{+}$ and $\Lambda(x^{0,-1})$.

(The relations in $\mathcal{E}^{\geq}$ yield a tensor product decomposition $\mathcal{E}^{\geq} = \Lambda(x^{0,-1}) \otimes \mathcal{E}^{+}$ (as a vector space).

The Hall algebra realization give rise to a geomenically defined coproduct Δ on $\mathcal{E}^{\geq}$ taking values in a suitably completed tensor product $\mathcal{E}^{\geq} \hat{\otimes} \mathcal{E}^{\geq}$.

When evaluated on $\Lambda(x^{m,n})$, the coproduct Δ on $\mathcal{E}^{\geq}$ has a leading term = standard coproduct on $\Lambda(x^{m,n})$:

Define:
- $\mathcal{E}^{+}_{<p} \subseteq \mathcal{E}^{+}$: subalgebra generated by $\Lambda(x^{m,n})$ for $\frac{n}{m} < p$
- $\mathcal{E}^{+}_{>p} \subseteq \mathcal{E}^{+}$: subalgebra generated by $\Lambda(x^{m,n})$ for $\frac{n}{m} > p$
- $\mathcal{E}^{\geq}_{<p} \subseteq \mathcal{E}^{\geq}$: subalgebra generated by $\mathcal{E}^{+}_{<p}$ and $\Lambda(x^{0,-1})$

$\ast$ $\mathcal{E}^{\geq}_{<p} = \Lambda(x^{0,-1}) \otimes \mathcal{E}^{+}_{<p}$.

Prop 5.1.1: For any $f \in \Lambda$ and coprime integers m,n with $m>0$ the coproduct in $\mathcal{E}^{\geq}$ evaluated on $f(x^{m,n})$ has the form

$$\Delta(f(x^{m,n})) = f[X^{m,n}_{(1)} + X^{m,n}_{(2)}] + (\text{terms in } \mathcal{E}^{\geq}_{<\frac{n}{m}} \hat{\otimes} \mathcal{E}^{\geq}_{>\frac{n}{m}}),$$

where $f[X^{m,n}_{(1)} + X^{m,n}_{(2)}] \in \Lambda(x^{m,n}) \otimes \Lambda(x^{m,n})$, and $X^{m,n}_{(1)}, X^{m,n}_{(2)}$ distinguish the tensor factors.

Recall that for any $\Gamma = \Gamma(\omega,y)$ satisfying $\frac{\Gamma(\omega,y)}{\Gamma(y,\omega)} = \Omega_1[-\frac{\omega}{y}\hat{M}] = \frac{(1-q+\frac{\omega}{y})(1-\frac{1}{q}\frac{\omega}{y})(1-\frac{1}{t}\frac{\omega}{y})}{(1-\frac{1}{q}+\frac{\omega}{y})(1-q\frac{\omega}{y})(1-t\frac{\omega}{y})}$, by Theorem 3.2.1, there is an isomorphism $\psi_\Gamma : \mathbf{S}_\Gamma \xrightarrow{\simeq} \mathcal{E}^{+}$.

Let $\mathbf{S}^{\geq}_\Gamma = \Lambda(x^{0,-1}) \otimes \mathbf{S}_\Gamma$ be the extended algebra isomorphic to $\mathcal{E}^{\geq}$ via

$$\psi^{\geq}_\Gamma : \mathbf{S}^{\geq}_\Gamma \xrightarrow{\simeq} \mathcal{E}^{\geq}$$

which is ψ_Γ on $\mathbf{S}_\Gamma$ and identity on $\Lambda(x^{0,-1})$. ($\psi^{\geq}_\Gamma = id \otimes \psi_\Gamma$)

$\therefore$ We have a coproduct Δ^Γ on $\Lambda(x^{0,-1}) \otimes \mathbf{S}_\Gamma$ corresponding under $\psi^{\geq}_\Gamma$ to the coproduct on $\mathcal{E}^{\geq}$.

indicates that these values are infinite sums of elements of different degrees in $(\Lambda(x^{0,-1}) \otimes \mathbf{S}^{k}_\Gamma) \otimes \mathbf{S}^{\ell-k}_\Gamma$

Negut gives a formula for the $\Delta^\Gamma_{k,\ell-k}$ of Δ^Γ with values in $(\Lambda(x^{0,-1}) \otimes \mathbf{S}^{k}_\Gamma) \hat{\otimes} \mathbf{S}^{\ell-k}_\Gamma$ when evaluated on $\mathbf{S}_\Gamma$:

Prop 5.2.1: Suppose $\Gamma(\omega,y)$ satisfy $\frac{\Gamma(\omega,y)}{\Gamma(y,\omega)} = \Omega_1[-\frac{\omega}{y}\hat{M}] = \frac{(1-q+\frac{\omega}{y})(1-\frac{1}{q}\frac{\omega}{y})(1-\frac{1}{t}\frac{\omega}{y})}{(1-\frac{1}{q}+\frac{\omega}{y})(1-q\frac{\omega}{y})(1-t\frac{\omega}{y})}$ and is a function of $\frac{\omega}{y}$. For any $H(z_1,\dots,z_\ell) \in \mathbf{S}^{\ell}_\Gamma$, we have

$$\Delta^\Gamma_{k,\ell-k}(H(z_1,\dots,z_\ell)) = \omega\,\Omega_1[-Y\hat{M}]\,x^{0,-1}\,\frac{H(\omega_1,\dots,\omega_k,y_1,\dots,y_{\ell-k})}{\prod_{i=1}^{k}\prod_{j=1}^{\ell-k}\Gamma(y_j,\omega_i)}$$

where $Y = y_1 + \dots + y_{\ell-k}$, and we distinguish the factors in $(\Lambda(x^{0,-1}) \otimes \mathbf{S}^{k}_\Gamma) \hat{\otimes} \mathbf{S}^{\ell-k}_\Gamma$ by writing elements of $\mathbf{S}_\Gamma$ as functions of $\omega_1,\dots,\omega_k$ in the first tensor factor, or $y_1,\dots,y_{\ell-k}$ in the second. Elements of $\Lambda(x^{0,-1})$ are understood to belong to the first tensor factor.

homogeneous

More precisely, let
- $(\Lambda(x^{0,-1}) \otimes \mathbf{S}^{k}_\Gamma)_d = (\psi^{\geq}_\Gamma)^{-1}((\mathcal{E}^{\geq})^{(k,d)})$ = subspace consisting of functions $h(x^{0,-1})f(\omega_1,\dots,\omega_k)$ homogeneous of degree d, where $\deg(h(x^{0,-1})) = -m$ where $m = \deg h$.
- $(\mathbf{S}^{\ell-k}_\Gamma)_d = \psi^{-1}_\Gamma((\mathcal{E}^{+})^{(\ell-k,d)})$ = subspace consisting of functions $g(y_1,\dots,y_{\ell-k})$ homogeneous of degree d.

Then $\Delta^\Gamma_{k,\ell-k}(H(z_1,\dots,z_\ell))$ = infinite sum with components in the spaces $(\Lambda(x^{0,-1}) \otimes \mathbf{S}^{k}_\Gamma)_{d_1} \otimes (\mathbf{S}^{\ell-k}_\Gamma)_{d_2}$ for some set of degrees with d_1 bounded above and d_2 bounded below.

The assumption on $\Gamma(\omega,y)$ ensures the R.H.S can be expanded as a formal Laurent series in ω_i^{-1} and y_j, multiplied by rational functions of $\frac{\omega_i}{\omega_j}$ and $\frac{y_i}{y_j}$. (Hence homogeneous of degree 0 in both ω and y).

$\therefore$ The component of $\Delta^\Gamma_{k,\ell-k}(H(z))$ in $(\Lambda(x^{0,-1}) \otimes \mathbf{S}^{k}_\Gamma)_{d_1} \otimes (\mathbf{S}^{\ell-k}_\Gamma)_{d_2}$ = the homogeneous component of these degree in the series expansion on R.H.S.

Suppose $\psi_\Gamma(H(z_1,\dots,z_\ell)) = f(x^{m,n})$ where $f \in \Lambda$ is homogeneous of degree d. Then $f(x^{m,n}) \in (\mathcal{E}^{+})^{(dm,dn)}$ and $H(z)$ is a function in $\ell = dm$ variables, homogeneous of degree dn.

$\therefore$ $\Delta^\Gamma_{k,\ell-k}(H(z_1,\dots,z_\ell))$ has components in $(\Lambda(x^{0,-1}) \otimes \mathbf{S}^{k}_\Gamma)_{d_1} \otimes (\mathbf{S}^{\ell-k}_\Gamma)_{d_2}$ for $d_1 + d_2 = dn$.

generated by $\Lambda(x^{a,b})$ with $f \in \Lambda$ of deg r s.t. $(ra,rb) = (k,d_1)$. Hence $\frac{b}{a} = \frac{d_1}{k}$

By Prop 5.1.1., the coproduct Δ on $f(x^{m,n})$ has components in $\mathcal{E}^{\geq}_{<\frac{n}{m}} \otimes \mathcal{E}^{+}_{>\frac{n}{m}}$. Since $(\Lambda(x^{0,-1}) \otimes \mathbf{S}^{k}_\Gamma)_{d_1} = (\psi^{\geq}_\Gamma)^{-1}((\mathcal{E}^{\geq})^{(k,d_1)})$, $\frac{d_1}{k} < \frac{n}{m}$ as the first tensor factor.

Similarly, $\frac{d_2}{\ell-k} > \frac{n}{m}$. Hence we have $d_1 < \frac{kn}{m}$ and $d_2 > \frac{(\ell-k)n}{m}$.

Also, the components contributing to the first term have $d_1 = \frac{kn}{m}$ and $d_2 = \frac{(\ell-k)n}{m}$, and necessarily with $m \mid k$.

Note that the first term in $\Delta(f(X^{m,n}))$ in Prop. 5.1.1 is in $\mathcal{E}^+ \otimes \mathcal{E}^+$, all terms in the RHS of Prop 5.2.1 contributing to it involve only the constant term in the factor $\omega\Omega[-\Upsilon\hat{H}X^{0,1}]$. We now have the following corollary from Prop. 5.1.1 and Prop 5.2.1:

Corollary 5.2.2: Suppose $\Gamma(\omega,y)$ satisfy $\frac{\Gamma(\omega,y)}{\Gamma(y,\omega)} = \Omega_\omega[-\frac{\omega}{y}\hat{H}] = \frac{(1-q\frac{\omega}{y})(1-\frac{1}{q}\frac{\omega}{y})(1-t\frac{\omega}{y})}{(1-\frac{1}{qt}\frac{\omega}{y})(1-q\frac{\omega}{y})(1-t\frac{\omega}{y})}$ and is a function of $\frac{\omega}{y}$. Let $H(z_1,\dots,z_\ell) \in S_\Gamma^\ell$.
Suppose $\Psi_\Gamma(H(z_1,\dots,z_\ell)) = f(X^{m,n})$ for some $f \in \Lambda$ with homogeneous degree d and (therefore) $\ell = dm$.
Let $0 \le e \le d$ and $k = em$, define

$$h(\omega,y) = h(\omega_1,\dots,\omega_k,y_1,\dots,y_{\ell-k}) = \left(\frac{H(\omega_1,\dots,\omega_k,y_1,\dots,y_{\ell-k})}{\prod_{i=1}^{k}\prod_{j=1}^{\ell-k}\Gamma(y_j,\omega_i)}\right)_{\max}$$

b/c $d_1 = \frac{kn}{m} = \frac{em\cdot n}{m} = en$

$d_2 = \frac{(\ell-k)n}{m} = \frac{(dm-em)n}{m} = (d-e)n.$

which is the homogeneous component of max. possible degree en in ω and minimum possible degree $(d-e)n$ in y.
Regard $h(\omega,y)$ as an element of $S_\Gamma^k \otimes S_\Gamma^{\ell-k}$, with $\omega_1,\dots,\omega_k$ in the first tensor factor and $y_1,\dots,y_{\ell-k}$ in the second. Then we have

$$(\Psi_\Gamma \otimes \Psi_\Gamma)(h(\omega,y)) = f[X_\omega^{m,n} + X_\omega^{m,n}]_{e,d-e} \in \Lambda(X^{m,n}) \otimes \Lambda(X^{m,n}),$$

where $f[X+Y]_{e,d-e}$ designates the homogeneous component of $f[X+Y]$ with degree e in X and $d-e$ in Y.